

EE 132: Midterm

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Problem 1

Taking the Laplace transform of both sides with zero initial conditions:

$$s^2Y(s) + 2sY(s) + 3Y(s) = 2sU(s) + 3U(s)$$

$$(s^2 + 2s + 3)Y(s) = (2s + 3)U(s)$$

$$\boxed{\frac{Y(s)}{U(s)} = \frac{2s + 3}{s^2 + 2s + 3}}$$

Problem 2

Let $E(s) = R(s) - \frac{2}{s+2}Y(s)$ denote the error signal. The two parallel forward paths sum to $(3 + s)E(s)$, which then passes through $\frac{1}{s+1}$:

$$Y(s) = \frac{s+3}{s+1}E(s) = \frac{s+3}{s+1} \left[R(s) - \frac{2}{s+2}Y(s) \right]$$

Multiplying through by $(s+1)(s+2)$:

$$Y(s)(s+1)(s+2) = (s+3)[(s+2)R(s) - 2Y(s)]$$

$$Y(s)[(s+1)(s+2) + 2(s+3)] = (s+2)(s+3)R(s)$$

$$Y(s)[s^2 + 5s + 8] = (s+2)(s+3)R(s)$$

$$\boxed{T(s) = \frac{Y(s)}{R(s)} = \frac{(s+2)(s+3)}{s^2 + 5s + 8}}$$

The characteristic polynomial $s^2 + 5s + 8$ has all positive coefficients, so the closed-loop system is stable. For a unit-step input $R(s) = 1/s$, the Final Value Theorem gives:

$$y_{ss} = \lim_{s \rightarrow 0} s \cdot T(s) \cdot \frac{1}{s} = T(0) = \frac{(2)(3)}{8} = \frac{6}{8}$$

$$\boxed{y_{ss} = \frac{3}{4}}$$

Problem 3

The open-loop transfer function is $G(s) = \frac{3}{s(s+4)}$ with unity feedback. The closed-loop transfer function is:

$$T(s) = \frac{G(s)}{1 + G(s)} = \frac{3}{s^2 + 4s + 3} = \frac{3}{(s+1)(s+3)}$$

Both poles are in the left half-plane, so the system is stable. For a unit-step input:

$$Y(s) = \frac{T(s)}{s} = \frac{3}{s(s+1)(s+3)}$$

Partial fraction decomposition:

$$\frac{3}{s(s+1)(s+3)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+3}$$

$$A = \left. \frac{3}{(s+1)(s+3)} \right|_{s=0} = \frac{3}{(1)(3)} = 1$$

$$B = \left. \frac{3}{s(s+3)} \right|_{s=-1} = \frac{3}{(-1)(2)} = -\frac{3}{2}$$

$$C = \left. \frac{3}{s(s+1)} \right|_{s=-3} = \frac{3}{(-3)(-2)} = \frac{1}{2}$$

$$y(t) = \left[1 - \frac{3}{2}e^{-t} + \frac{1}{2}e^{-3t} \right] u(t)$$

Problem 4

Stability range. The open-loop transfer function is:

$$G(s) = \left(k_1 + \frac{k_2}{s} \right) \cdot \frac{1}{(s-1)(s+3)} = \frac{k_1s + k_2}{s(s-1)(s+3)}$$

The closed-loop characteristic equation is $1 + G(s) = 0$, i.e. $s(s-1)(s+3) + k_1s + k_2 = 0$. Expanding:

$$s^3 + 2s^2 + (k_1 - 3)s + k_2 = 0$$

Routh array:

$$\begin{array}{c|cc} s^3 & 1 & k_1 - 3 \\ s^2 & 2 & k_2 \\ s^1 & \frac{2(k_1-3)-k_2}{2} & 0 \\ s^0 & k_2 & \end{array}$$

For all first-column entries to be positive:

$$\boxed{k_2 > 0 \quad \text{and} \quad 2k_1 - k_2 > 6}$$

System type. The open-loop transfer function has exactly one pole at the origin, so the closed-loop system is **Type 1**.

Steady-state error, $k_1 = 5$, $k_2 = 1$. Check stability: $k_2 = 1 > 0$ and $2(5) - 1 = 9 > 6$. The system is stable.

For a Type 1 system with a step input, the position error constant is $K_p = \lim_{s \rightarrow 0} G(s) \rightarrow \infty$, so:

$$\boxed{e_{ss} = \frac{1}{1 + K_p} = 0}$$

Problem 5

Closed-loop transfer function. The open-loop gain is $G(s) = \frac{k_1}{(s+k_2)(s-1)}$, giving:

$$T(s) = \frac{k_1}{(s+k_2)(s-1) + k_1} = \frac{k_1}{s^2 + (k_2 - 1)s + (k_1 - k_2)}$$

Comparing to the standard second-order form $\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$:

$$\omega_n^2 = k_1 - k_2, \quad 2\zeta\omega_n = k_2 - 1$$

Extract σ from the 1% settling time.

$$t_s = \frac{4.6}{\zeta\omega_n} = 9.2 \text{ s} \implies \zeta\omega_n = \frac{4.6}{9.2} = 0.5$$

Therefore $k_2 - 1 = 2(\zeta\omega_n) = 1$, so $\boxed{k_2 = 2}$.

Extract ω_n from the rise time.

$$t_r = \frac{1.8}{\omega_n} = 1.8 \text{ s} \implies \omega_n = 1 \text{ rad/s}$$

Therefore $k_1 - k_2 = \omega_n^2 = 1$, so $\boxed{k_1 = 3}$.

Verification. With $k_1 = 3$, $k_2 = 2$:

$$T(s) = \frac{3}{s^2 + s + 1}, \quad \zeta = \frac{k_2 - 1}{2\omega_n} = \frac{1}{2} = 0.5$$

Both poles lie in the open left half-plane ($0 < \zeta < 1$, underdamped), confirming the system is stable and the specs are achievable.